The Invisible Ledger: Quantifying ASEAN’s Unmeasured Platform Economy

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Abstract

This paper estimates an administrative information gap between platform-mediated transaction value and the revenue booked by the platforms that coordinate it. Using Grab, GoTo, and Shopee disclosures, I obtain a $66.8 billion Indonesia-focused estimate of unbooked gross transaction value in 2023, corresponding to an ecosystem ratio of 16.0×. The GoTo Group time series, rebuilt from primary filings and contemporaneous annual exchange rates, shows an unbooked share persistently between 97.6% and 98.9% over 2020–2023 rather than a monotonic increase. Scope-matched historical Grab and Shopee disclosures show similarly large transaction-to-revenue gaps. Applying the base estimate's 5.88% blended retained share to four additional ASEAN markets yields an order-of-magnitude scenario of approximately $200 billion. Malaysia's 2024 low-value-goods regime collected about US$104 million (RM476 million as reported); a historical-FX before-and-after comparison with Vietnam is about US$136 million. The policy implication is reporting first, liability second.

Keywords: digital economy; platform economy; fiscal capacity; informal sector; third-party reporting; ASEAN; tax administration.

JEL codes: H25; H26; O17; O33; F23.

1. Introduction

Southeast Asia’s digital economy grew from $98 billion in 2020 to $263 billion in 2024 (Google, Temasek and Bain 2024). Platform scale and tax-administrative visibility are not the same thing. A platform records each transaction, determines participant payouts, and accounts for its own revenue, while the gross receipts earned by participating restaurants, delivery providers, or vendors may remain outside routine third-party income reporting. I call this administrative information gap the “invisible wedge.” The empirical question is whether the digitally recorded but separately unreported flow is large enough to matter for tax administration.

This paper evaluates the invisible wedge in the Association of Southeast Asian Nations (ASEAN) using an Indonesia-focused three-platform estimate and then calibrating that base to four additional markets: Vietnam, the Philippines, Thailand, and Malaysia.

The term invisible refers to the administration’s information channel, not to criminal concealment. The transactions are digitally recorded by platform operators; however, these records are generally used for payment processing and platform accounting purposes rather than being transmitted as formal income reports for participating businesses and workers, such as restaurants, delivery providers, and vendors. A schematic example clarifies the accounting. Suppose a platform processes 100 units of transaction value and books 10 as revenue. The remaining 90 is visible in the platform’s transaction system but is not platform revenue. It may include merchant receipts, driver payouts, inventory costs, taxes already paid elsewhere, and other pass-through values.

I evaluate the invisible wedge using an ecosystem ratio—the gross transaction value (GTV) not booked as platform revenue relative to the revenue the platform does book—constructed from company financial disclosures. The core cross-platform estimate uses fiscal-year 2023 data for Grab, GoTo, and Shopee. Grab and Shopee inputs are Indonesia-specific or transparently derived for Indonesia; GoTo reports both relevant inputs directly only at Group level, and its 2023 Group also includes operations outside Indonesia. The resulting $66.8 billion figure is therefore an Indonesia-focused base estimate rather than a perfectly country-isolated measurement. The four non-Indonesian estimates are scenario calibrations rather than independent country measurements.

I focus on Grab, GoTo, and Shopee (operated by Sea Limited) because they are large ASEAN-origin consumer transaction platforms that directly intermediate payments between consumers and merchants or service providers and publish financial disclosures from which transaction value and platform revenue can be observed or transparently derived. Their business models therefore place the intermediary that records the transaction and computes the payout at the center of the paper’s third-party-reporting question; GoTo is especially informative because its pre-2024 disclosures are concentrated in Indonesia.

The paper therefore measures gross participant-facing transaction value, not participant profit or unpaid tax.

The regional figure is therefore an order-of-magnitude scenario anchored on the Indonesia-focused base estimate, not five independent country measurements.

1.1 Contributions

Overall, my study makes several important contributions. First, I provide directly reported and transparently derived platform-level inputs for an Indonesia-focused 2023 base estimate, then use its blended retained share to calibrate four additional ASEAN markets.

Only GoTo reports both relevant inputs independently, but at Group rather than country-isolated level; every derived or scope-mismatched input is labelled by provenance.

Second, I use Malaysia’s January 2024 low-value-goods expansion as a descriptive policy benchmark. I report the policy’s directly observed collections alongside a two-country before-and-after comparison with Vietnam, without a significance claim or causal interpretation.

Third, I rebuild GoTo's 2020–2023 Group series from primary filings and show a persistently large transaction-to-net-revenue gap: the unbooked share remains between 97.6% and 98.9% rather than rising monotonically. This within-company evidence is less dependent on the cross-platform calibration used elsewhere.

Fourth, I use historical scope-matched Grab and Shopee disclosures to test whether large transaction-to-revenue gaps persist when numerator and denominator are aligned more closely within platform segments. Finally, an exploratory earnings-announcement analysis does not show a stable pricing relationship in the current sample.

1.2 What this paper does and does not claim

Table 1 states the inferential boundaries before the estimates are presented. The correction history is retained in Appendix A so that provenance is auditable without turning the main argument into an errata list.

Table 1. Claims supported—and not supported—by the paper’s evidence

Established here	Not established here
An order-of-magnitude regional estimate built from one Indonesia-focused three-platform base and four calibrated rescalings.	Five independent country measurements or a complete measure of ASEAN’s informal economy.
Two descriptive Malaysia benchmarks: about US$104M of directly reported LVG collections (RM476M as reported; converted at 2024 period-average FX) and an approximately US$136M historical-FX before-and-after difference.	A causal effect identified by matched tax bases, parallel trends, or a statistically informative panel.
A primary-source GoTo Group series in which the unbooked share remains between 97.6% and 98.9% over 2020–2023.	That the GoTo Group series is perfectly country-isolated or that its unbooked share rose monotonically.
The wedge measures transaction value not booked as platform revenue.	That all participant-facing GTV is profit, taxable income, unpaid tax, or GDP value-added.
No stable announcement-window pricing relationship is detected in the current exploratory investor sample.	That the ecosystem ratio has no investor relevance or that the null is precisely estimated.

2. Literature review and theoretical framework

2.1 Platform economics and the location of value

The economics of multisided platforms begins with the interaction of distinct user groups whose participation affects the value received by the other side. Rochet and Tirole (2003), Caillaud and Jullien (2003), Parker and Van Alstyne (2005), and Armstrong (2006) show why platform pricing cannot be read as a simple markup over cost: prices and commissions are chosen to balance cross-group network effects, participation, and competition. Hagiu and Wright (2015) further distinguish a multisided platform from a vertically integrated firm by locating decision rights and entrepreneurial effort with affiliated participants rather than inside the platform. Evans and Schmalensee (2016) translate the same point into the business-model literature. These results explain why low retained shares can coexist with large ecosystems and why platform revenue is a poor proxy for the transaction value coordinated through the platform.

2.2 Informality, platform work, and measurement

Traditional informal-economy estimates rely on household surveys, firm surveys, or indirect macroeconomic indicators. Medina and Schneider (2019) estimate a large global shadow economy; La Porta and Shleifer (2014) emphasise the small scale and low productivity of informal firms; and Ulyssea (2018) shows that registration and off-the-books employment are distinct margins whose welfare effects cannot be inferred from formal status alone. The International Labour Organization (2021, 2023) likewise treats platform work and informality as overlapping but non-identical categories.

Platform participants occupy an unusual position in this literature. Their transactions are priced, recorded, and settled through a formal intermediary, but the participant may remain self-employed, unregistered, below filing thresholds, or simply outside an automatic reporting channel. The object measured here is therefore not the shadow economy as conventionally defined. It is a digitally recorded flow whose administrative treatment resembles self-reported income even though an intermediary already possesses the record.

2.3 Third-party reporting as tax capacity

A central result in public finance is that tax compliance depends strongly on third-party information and remittance structure. Kleven et al. (2011) find an evasion rate of 0.3 percent on income subject to third-party reporting versus 37 percent on self-reported income in Denmark. Pomeranz (2015) shows that the VAT paper trail creates self-enforcement across firms, while Naritomi (2019) demonstrates that consumer-held records can raise reported sales. Kleven, Kreiner and Saez (2016) formalise firms as fiscal intermediaries, and Slemrod (2019) places information reporting and remittance regimes among the core instruments of modern enforcement.

Jensen (2022) connects this mechanism to development: as employment moves into organisations that keep records and report income, the feasible tax base expands. Platform work creates a new configuration - a large intermediary with detailed records but, in many jurisdictions, no comprehensive obligation to report seller or worker income. Collins et al. (2019) and Garin et al. (2023) show how platform information returns affect the measured prevalence of gig work in the United States. The administrative problem is therefore not a lack of data generation; it is the absence, incompleteness, or fragmentation of the reporting rule.

Two finance studies extend this administrative-data channel. Barrios, Hochberg, and Yi (2022) show that gig-platform entry increases new business formation and small-business lending, while Denes, Lagaras, and Tsoutsoura (2025) use U.S. tax-return microdata to show that gig work can serve as a pathway into entrepreneurship. Together, these studies reinforce the idea that platform-mediated work leaves economically meaningful financial and administrative records.

2.4 Digital taxation and platform-reporting rules

Corporate digital-tax reforms and platform-participant reporting address different bases. The OECD Pillar One blueprint concerns the allocation of taxing rights over large multinational enterprises (OECD 2020b). By contrast, the OECD's Forum on Tax Administration report and Model Rules require platforms to collect and report information on income realised by sellers and service providers (OECD 2019, 2020a). The European Union's DAC7 regime similarly imposes due-diligence and reporting duties on platform operators (European Union 2021). These instruments provide the closest policy precedent for the mechanism studied here: using the platform's existing transaction record to improve participant-level visibility without treating platform revenue as the participant tax base.

3. Theoretical framework: the ecosystem ratio

Return to the schematic from the Introduction: if a platform processes 100 units of transaction value and books 10 as revenue, 90 units pass through the ecosystem without being recorded as the platform's own revenue. The ecosystem ratio formalizes that comparison: (100 − 10) / 10 = 9.

$$\textbf{Ecosystem ratio} = \textbf{(GTV - platform revenue) / platform revenue} \quad \textbf{(1)}$$

Gross transaction value (GTV) is the total value of transactions processed through a platform, regardless of who ultimately receives the proceeds. This construction is consistent with the platform-economics literature showing that booked platform revenue need not track the total transaction value coordinated through a multisided platform (Rochet and Tirole 2003; Evans and Schmalensee 2016).

The ecosystem ratio describes the scale of participant-facing transaction value relative to the platform's auditable revenue base. A higher ratio indicates a larger unbooked transaction flow relative to the platform's own revenue; it does not by itself imply greater participant profit, taxable income, value-added, non-compliance, or tax due.

This distinction is essential. Platform filings and investor disclosures can make the accounting gap measurable, but the tax treatment of the residual depends on participant type, deductible costs, thresholds, registration status, and taxes already remitted through other channels. The empirical contribution is to quantify the size of the recorded flow that current corporate-level digital taxes do not directly reach.

4. Data, methodology, and provenance

4.1 Institutional background: Grab, GoTo, and Shopee

I focus on Grab, GoTo, and Shopee (operated by Sea Limited) because they are large ASEAN-origin consumer platforms that directly intermediate transactions and provide public financial disclosures sufficient to observe or transparently derive transaction value and platform revenue. Google, Temasek and Bain (2024) estimate that Southeast Asia's digital economy grew from $98 billion in 2020 to $263 billion in 2024. Public listing makes these three platforms unusually observable relative to private competitors, which is essential for the company-level measurement used here.

Grab Holdings Limited. Grab began as a ride-hailing app in Malaysia in 2012 and has since expanded into a multi-service "super-app" spanning Mobility (ride-hailing), Deliveries (food and grocery delivery), and Financial Services (digital payments, lending, and insurance) across eight Southeast Asian countries. It listed on Nasdaq in December 2021 through a special-purpose acquisition company (SPAC) merger with Altimeter Growth Corp. Grab reports "On-Demand GMV," where gross merchandise value (GMV) refers here to the combined transaction value of Mobility and Deliveries; Financial Services is reported separately and excluded from this measure.

PT GoTo Gojek Tokopedia Tbk (GoTo). GoTo is the product of a 2021 merger between Gojek (ride-hailing and on-demand services, founded 2010) and Tokopedia (e-commerce, founded 2009), Indonesia's two largest homegrown technology companies. It listed on the Indonesia Stock Exchange in April 2022. Indonesia is GoTo's core market, but its 2023 Group structure also includes operating entities outside Indonesia, including Singapore and Vietnam. Group GTV and net revenue are therefore not a country-level Indonesia split. In January 2024, GoTo combined Tokopedia's e-commerce operations with TikTok Shop, a transaction in which TikTok's parent, ByteDance, committed over $1.5 billion of investment; Tokopedia was deconsolidated to an associate-company basis from that point forward. This is a structural break in GoTo's own reported GTV and revenue series, not a mere reporting change—pre- and post-2024 figures are not on a like-for-like consolidation basis and should not be pooled without adjustment.

Sea Limited / Shopee. Sea Limited, headquartered in Singapore, operates three segments: Shopee (e-commerce), Garena (digital entertainment), and Monee (formerly SeaMoney; digital financial services). Sea listed on the NYSE in October 2017, and Shopee operates across Southeast Asia, Taiwan, and Latin America. Where Shopee segment revenue is unavailable, pairing Shopee GMV with Sea's consolidated revenue introduces a scope mismatch because consolidated revenue also includes Garena and Monee; the historical robustness analysis therefore uses Shopee segment revenue wherever the filings permit.

Scope note. These three platforms are not interchangeable observations of one phenomenon. Grab and GoTo are dominantly ride-hailing/on-demand businesses with e-commerce or fintech as secondary lines; Shopee is dominantly e-commerce with gaming and fintech as secondary lines. Grab and Shopee are multi-country, while GoTo is Indonesia-dominant but its Group disclosures are not perfectly country-isolated. This heterogeneity is why Section 7 treats the 2023 base as Indonesia-focused and the other four ASEAN markets as calibrated rescalings rather than presenting all five as independently observed.

4.2 Sample construction

I obtain platform revenue and gross transaction value from each company's primary financial disclosures: Grab and Sea Limited Form 20-F filings and GoTo annual and quarterly reports, supplemented by Momentum Works (2024) only where an Indonesia-specific market-share input is not separately disclosed. The core cross-platform sample is fiscal year 2023. Table 2 distinguishes directly reported figures, derived Indonesia inputs, and GoTo's directly reported but Group-level inputs.

Take rate is platform revenue divided by GTV. The base estimate's blended take rate is the same ratio computed across all three platform rows combined—total platform revenue divided by total GTV, 5.88%—and is used only for the four-country scenario calibration in Section 7. Because the GoTo row is Group-level, I refer to this as the Indonesia-focused blended take rate rather than a perfectly country-isolated Indonesia parameter.

For robustness, a separate historical sample uses primary filings to extend Grab segment data through 2019–2021 and Shopee GMV/segment revenue through 2017–2021. These historical observations are company- or segment-level rather than Indonesia-specific and are reported separately so that geographic and accounting scopes are not mixed with Table 3.

Table 2. Source, derivation, and geographic scope of each 2023 Indonesia-focused platform input

Platform	Metric	Source	Provenance
Grab	Revenue	Grab 2023 Form 20-F	Directly disclosed at country level.
Grab	GTV	Derived	Indonesia revenue divided by company-wide take rate (11.24%); country GTV is not independently disclosed.
GoTo Group	GTV and revenue	GoTo FY2023 annual report	Both inputs directly reported at Group level; the Group also operated outside Indonesia, and no country-level GTV split is disclosed.
Shopee	GTV and revenue	Sea 2023 Form 20-F; Momentum Works 2024	Regional revenue/GMV disclosed; Indonesia GMV estimated from market share, with revenue derived from disclosed take rate (10.06%).

Only GoTo supplies both inputs independently, but its reported figures are Group-level. Grab and Shopee ratios inherit the take-rate assumptions used to derive one or both Indonesia-specific components.

4.3 Indonesia-focused platform estimates

Table 3 reports the three platform rows used for the 2023 Indonesia-focused base estimate. GoTo's markedly higher ecosystem ratio (40.0×) relative to Grab and Shopee (roughly 8–9×) reflects its consolidated GTV reporting: Tokopedia's full marketplace GMV is included while GoTo books only a thin commission as revenue. The spread across platforms therefore reflects reporting convention as well as economic structure and should not be read as evidence that the ratio is a stable cross-platform parameter. The combined 5.88% blended take rate provides the calibration basis for Section 7, subject to the geographic-scope qualification on the GoTo row.

Table 3. Indonesia-focused platform GTV, net revenue, unbooked GTV, and ecosystem ratio, FY2023

Note: Grab and Shopee rows are Indonesia-specific or transparently derived for Indonesia. GoTo uses directly reported Group GTV and net revenue because no country-level GTV split is disclosed; the GoTo Group also operated outside Indonesia in 2023. Table 3 is therefore Indonesia-focused rather than a perfectly country-isolated sum. GoTo's 2023 rupiah inputs are converted at the 2023 period-average official IDR/USD rate.

Platform	GTV	Net revenue	Unbooked GTV	Ecosystem ratio
Grab	$5.40B	$0.61B	$4.80B	7.9×
GoTo Group	$39.81B	$0.97B	$38.84B	40.0×
Shopee	$25.80B	$2.60B	$23.20B	8.9×
Total	$71.00B	$4.18B	$66.83B	16.0×

Wedges and ratios are calculated from unrounded source values. Displayed components are rounded and may therefore not reproduce the reported results exactly.

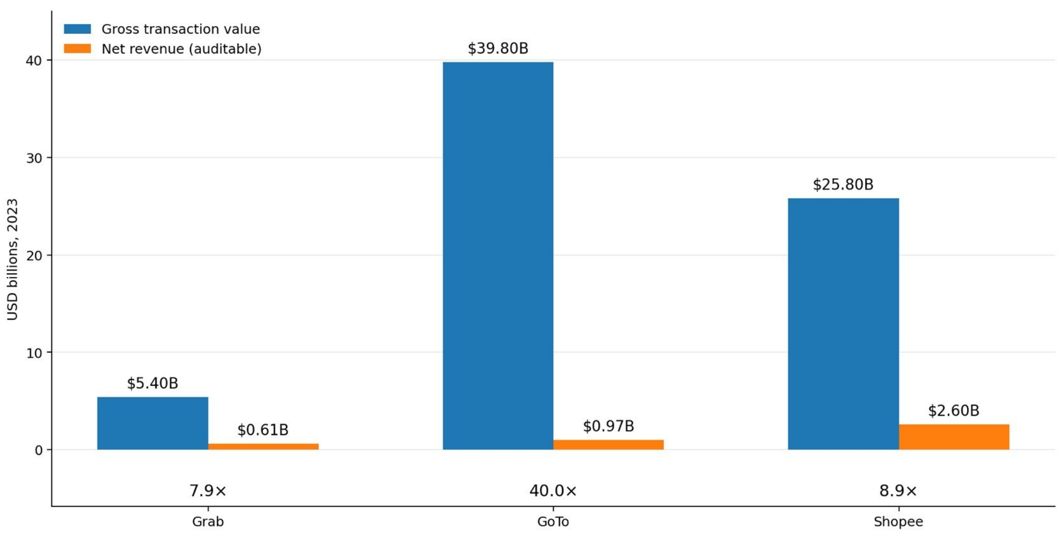

Figure 1. Indonesia-focused platform GTV and net revenue by platform, FY2023. The ecosystem ratio equals (GTV − platform revenue) / platform revenue; cross-platform differences partly reflect reporting conventions. The GoTo bar uses Group-level figures; see Table 3 note.

Table 3 is deliberately single-year and Indonesia-focused because public filings do not support a consistent multi-year, country-isolated panel for all three platforms. In Grab's 2019–2021 geographic disclosures, Indonesia is grouped with Vietnam, Myanmar, and Cambodia under "Rest of Southeast Asia," so Indonesia-specific historical revenue cannot be isolated from the company's filings. GoTo's historical headline GTV and net revenue are Group-level. To answer the time-series question without pretending these scopes are identical, Table 3B reports a separate company/segment-level robustness sample constructed from primary filings.

Table 3B. Scope-matched historical ecosystem ratios from primary platform filings

Platform / scope	Year	GMV/GTV (USD B)	Matched revenue (USD B)	Ecosystem ratio	Provenance
Shopee e-commerce	2017	4.113	0.009	454.3×	Shopee GMV + segment revenue
Shopee e-commerce	2018	10.279	0.270	37.1×	Shopee GMV + segment revenue

Platform / scope	Year	GMV/GTV (USD B)	Matched revenue (USD B)	Ecosystem ratio	Provenance
Shopee e-commerce	2019	17.576	0.834	20.1×	Shopee GMV + segment revenue
Shopee e-commerce	2020	35.400	2.167	15.3×	Shopee GMV + segment revenue
Shopee e-commerce	2021	62.600	5.123	11.2×	Shopee GMV + segment revenue
Grab Mobility + Deliveries	2019	8.662	−0.629	n/m	Negative segment revenue after incentive netting
Grab Mobility + Deliveries	2020	8.700	0.443	18.6×	Scope-matched segment GMV + revenue
Grab Mobility + Deliveries	2021	11.317	0.604	17.7×	Scope-matched segment GMV + revenue

Note: Ratios use (GMV/GTV − matched revenue) / matched revenue and are calculated from unrounded source values. Shopee figures use e-commerce GMV and Shopee segment revenue from Sea Limited SEC filings; Grab figures use Mobility + Deliveries GMV and revenue from Grab's FY2021 Form 20-F. The 2019 Grab ratio is not meaningful because segment revenue is negative after incentive netting. These observations are not Indonesia-specific and do not replace Table 3.

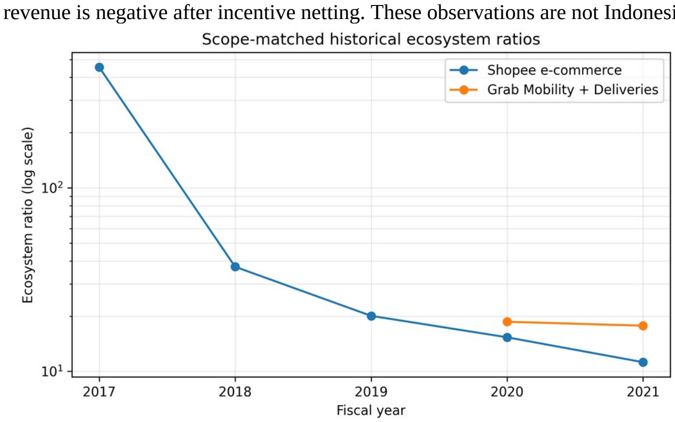

Figure 1B. Scope-matched historical ecosystem ratios for Shopee e-commerce and Grab Mobility + Deliveries. The y-axis is logarithmic; Grab 2019 is omitted because matched segment revenue is negative. The exhibit is company/segment-level robustness evidence, not an Indonesia time series.

To make the 2022–2025 extension explicit without conflating it with the Indonesia-focused base estimate, Table 3C reports the year-by-platform ecosystem ratio from the source-verified company-wide investor-panel observations. For each platform-year, the ratio is calculated from the summed GMV/GTV and revenue of the quarters actually observed in that year; partial-year rows are not annualized and missing quarters remain missing. Figure 1C visualizes the same annual or partial-year ratios. These exhibits are descriptive within-platform extensions, not a multi-year Indonesia panel or a cross-platform ranking.

Table 3C. Extended company-wide ecosystem-ratio evidence by year and platform, 2022–2025

Year	Platform	Observed quarters	Ecosystem ratio	Coverage / scope note
2022	Grab	4/4 (Q1–Q4)	13.0×	company-wide; full-year observed
2022	Sea/Shopee	4/4 (Q1–Q4)	4.9×	Shopee GMV vs Sea revenue scope used in investor panel; full-year observed
2023	Grab	2/4 (Q1, Q3)	8.0×	company-wide; partial year, not annualized

Year	Platform	Observed quarters	Ecosystem ratio	Coverage / scope note
2023	GoTo	2/4 (Q3, Q4)	38.8×	company/group reporting scope; partial year, not annualized
2023	Sea/Shopee	2/4 (Q1, Q2)	4.7×	Shopee GMV vs Sea revenue scope used in investor panel; partial year, not annualized
2024	Grab	4/4 (Q1–Q4)	5.6×	company-wide; full-year observed
2024	GoTo	2/4 (Q1, Q4)	34.8×	post-Tokopedia-deconsolidation reporting regime; partial year, not annualized
2024	Sea/Shopee	1/4 (Q4)	4.7×	Shopee GMV vs Sea revenue scope used in investor panel; single observed quarter
2025	Grab	4/4 (Q1–Q4)	5.6×	company-wide; full-year observed
2025	GoTo	4/4 (Q1–Q4)	36.7×	post-deconsolidation reporting regime; full-year observed
2025	Sea/Shopee	4/4 (Q1–Q4)	5.9×	Shopee GMV vs Sea revenue scope used in investor panel; full-year observed

Note: The ecosystem ratio equals (GMV/GTV − revenue) / revenue and is calculated from the summed observed quarters within each platform-year. Rows with fewer than four quarters are partial-year observations and are not annualized. Missing quarters remain missing. GoTo 2022 is omitted because the prior source-lineage file derives 2022Q1 as first-half 2022 minus directly reported 2022Q2; that derived quarter is not treated as a standalone public-announcement observation. Cross-platform ratio levels are not economically comparable because reporting scope and monetization conventions differ; the intended comparison is within platform over time.

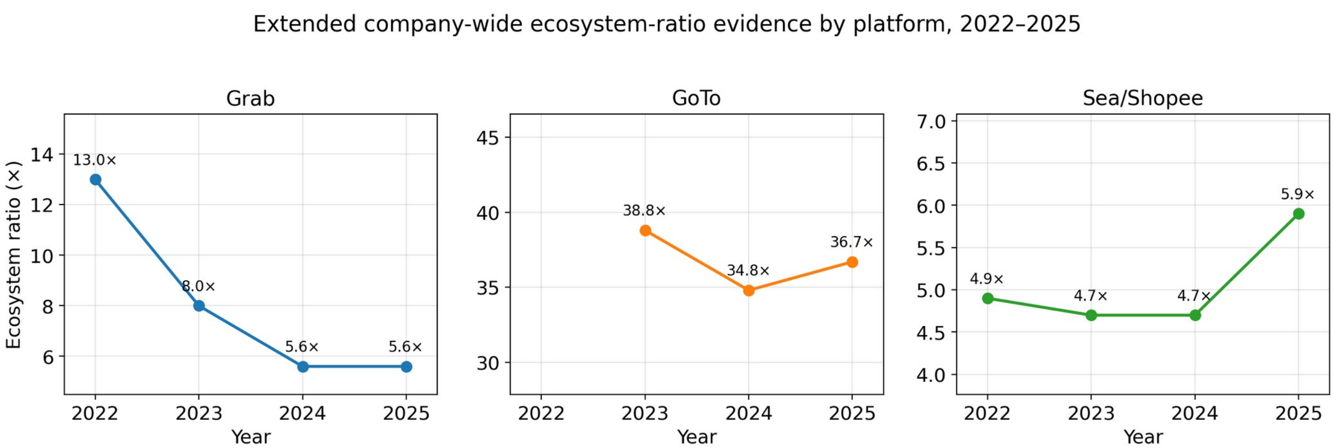

Figure 1C. Extended company-wide ecosystem-ratio evidence by platform, 2022–2025. Points correspond to the annual or partial-year ratios reported in Table 3C; missing platform-years are left blank rather than interpolated. Each platform should be read on its own scale because ratio levels reflect different reporting and monetization conventions. GoTo's 2024–2025 observations follow the Tokopedia deconsolidation and are not treated as directly comparable with its pre-2024 reporting regime.

4.4 Sensitivity of the Indonesia-focused estimate

Table 4 varies the Indonesia-focused blended take rate from 5% to 10% while holding the base estimate's total GTV fixed. Grab's and Shopee's Indonesia figures also depend on platform-specific take rates and, for Shopee, an estimated market share; those additional sources of uncertainty are not propagated in this sensitivity check.

Table 4. Sensitivity of the Indonesia-focused unbooked GTV and ecosystem ratio to the assumed blended take rate

Take rate	Platform revenue	Unbooked GTV	Ecosystem ratio
5.00% (low)	$3.55B	$67.45B	19.0×
5.88% (base)	$4.18B	$66.83B	16.0×
7.50% (earlier draft)	$5.33B	$65.68B	12.3×
10.00% (high)	$7.10B	$63.90B	9.0×

The dollar wedge is comparatively insensitive to the take rate because the retained share is small; the ratio is largely a transformation of that assumed share.

4.5 The investor-response sample

The pricing test in Section 8 draws on a 35-firm-quarter panel covering Grab (14 quarters), GoTo (9 quarters), and Sea (12 quarters), 2022Q1 through 2026Q1. Every GMV, revenue, and announcement-date figure is read directly from the platform's own quarterly earnings release, with a source citation carried alongside each value; none is interpolated or back-filled. Table 3C and Figure 1C report the requested 2022–2025 extension by year and platform, while Table 5 reports the full panel's coverage and platform-level descriptive statistics.

Coverage is uneven because a quarter enters the investor-response panel only when GMV/GTV, revenue, and the earnings-announcement date are all source-verified from public disclosures; missing quarters are left missing rather than interpolated.

Table 5. Investor-response sample coverage and platform-level descriptive statistics, 2022Q1–2026Q1

Panel A. Number of firm-quarter observations by year and platform

Year	GRAB	GOTO	SEA	Total
2022	4	0	4	8
2023	2	2	2	6
2024	4	2	1	7
2025	4	4	4	12
2026	0	1	1	2
Total	**14**	**9**	**12**	**35**

Panel B. Ecosystem-ratio distribution by platform

Platform	N	Ratio mean	Ratio median
GRAB	14	9.35	6.69
GOTO	9	38.45	37.58
SEA	12	6.25	5.90

Note: Ecosystem ratios are unit-free but not economically comparable across platforms because reporting scope differs. Sea pairs Shopee GMV with consolidated revenue where segment revenue is unavailable, while GoTo's marketplace GTV follows a different monetization convention. Monetary GMV/revenue descriptives are omitted because FX conversion would standardize currency, not economic scope; monetary values elsewhere are reported in U.S. dollars (USD), with original local-currency amounts retained where needed for provenance. Section 8 therefore treats pooled ratio levels as descriptive and emphasizes within-platform changes.

4.6 Reconciling the Indonesia and company-wide ecosystem ratios

The paper's existing Indonesia-specific measurement (Table 3) and the company-wide panel built for the pricing test (Table 5) report different ecosystem ratios for Grab, and the difference is large enough to require explanation rather than a footnote. Table 3 puts Grab's Indonesia ecosystem ratio at 7.9× ($4.80B unbooked GTV against $605M revenue in 2023, using the paper's own (GTV−Revenue)/Revenue definition). The company-wide panel shows Grab's ratio settling into a stable 6.3–6.7× band across all eight quarters of 2024–2025.

These are not competing estimates of the same quantity, and reporting them side by side without explanation invites a reader to assume one is wrong. Three things separate them.

Scope. The Indonesia figure isolates one country; the panel figure is Grab's entire multi-country operation across all eight markets it serves. A country-specific ratio differing from the company-wide average is therefore an expected consequence of aggregating across markets, not necessarily an anomaly requiring a separate explanation.

Derivation versus direct observation. The Indonesia figure is derived, not disclosed: Grab reports Indonesia revenue directly in its 20-F, but not Indonesia-specific GTV, so the $5.40B figure is backed out by dividing disclosed revenue by Grab's disclosed company-wide take rate (11.24%). That construction imports the company-wide take rate into a country-specific estimate, which mechanically pulls the Indonesia ratio toward some average behavior even as it tries to isolate one country—a source of measurement noise the company-wide panel does not have, since GTV and revenue are both read directly off the same release for every observation.

What an investor can actually see. This is the sharper point for the pricing test specifically. No public filing gives an investor an Indonesia-only ecosystem ratio for Grab—it does not exist as a disclosed number anywhere. What a market participant observes, quarter over quarter, is the company-wide figure: the one that enters the panel in Section 8. The Indonesia estimate is the right measurement for this paper's institutional argument about wedge size; the company-wide figure is the right measurement for the pricing argument about what investors can respond to. Using each ratio for the question it actually answers, rather than treating them as two attempts at one number, resolves the apparent conflict.

5. GoTo results: persistence of a large transaction–revenue gap

Table 6 gives a within-company time series using GoTo's primary Group disclosures. The 2020–2021 rows use the company's pro forma performance highlights as if Tokopedia had been consolidated from 1 January 2020; 2022–2023 are reported Group figures. Because GoTo also operated outside Indonesia, these observations are Group-level rather than country-isolated Indonesia figures.

Table 6. GoTo Group GTV, net revenue, unbooked GTV, and unbooked share, 2020–2023 (2020–2021 pro forma)

Note: Source values are Rp330.176T/Rp4.820T (2020), Rp461.602T/Rp5.156T (2021), Rp613.362T/Rp11.349T (2022), and Rp606.547T/Rp14.785T (2023) for GTV/net revenue. USD values apply each year's World Bank WDI/IMF period-average IDR/USD rate (World Bank n.d.) (14,582.2; 14,308.1; 14,849.9; 15,236.9). The unbooked share is calculated in rupiah before conversion and is therefore currency-invariant.

Year	GTV	Net revenue	Unbooked GTV	Share unbooked
2020	$22.64B	$0.33B	$22.31B	98.5%
2021	$32.26B	$0.36B	$31.90B	98.9%
2022	$41.30B	$0.76B	$40.54B	98.1%
2023	$39.81B	$0.97B	$38.84B	97.6%

The share of GTV not booked as GoTo net revenue remains very high throughout the primary-source series, ranging from 97.6% to 98.9%. It does not rise monotonically: it peaks in 2021 and declines to 97.6% in 2023 as net revenue grows relative to GTV. The finding is therefore persistence of a large transaction–revenue gap, not evidence that the gap widened over 2020–2023.

6. Malaysia's 2024 base expansion

Malaysia extended its digital tax base in January 2024 through the Low Value Goods (LVG) regime (Royal Malaysian Customs Department 2023; Ministry of Finance, Malaysia 2023). Two descriptive benchmarks are available: the LVG regime's directly reported collections and a two-country before-and-after comparison with Vietnam. They answer different questions and are therefore reported separately rather than combined into a single treatment-effect estimate.

The direct reading is the policy's own reported collection. At the 2024 period-average USD/MYR rate, Malaysia's reported Low Value Goods (LVG) tax revenue is about US$104 million (RM476 million as reported), alongside about US$354 million of pre-existing Sales Tax on Digital Services (SToDS) receipts (RM1.62 billion as reported). The original ringgit amounts remain the source-of-record values; the U.S.-dollar amounts are this paper's contemporaneous historical-FX conversions.

The comparative reading is a descriptive two-country before-and-after comparison with Vietnam's foreign-supplier e-portal collections. Under the paper's contemporaneous historical-FX convention, Malaysia's SToDS receipts are about US$252.1 million in 2023 (RM1.15 billion as reported) and about US$354.0 million in 2024 (RM1.62 billion as reported); adding 2024 LVG receipts gives about US$458.0 million (RM2.096 billion as reported) for that year. Vietnam's foreign suppliers declared and paid about US$289.9 million in 2023 (VND6.896 trillion as reported) and about US$359.5 million in 2024 (VND8.687 trillion as reported) through the e-portal (VietnamPlus 2023; General Department of Taxation Vietnam 2024). The parenthetical local-currency amounts are retained as the source-of-record values used for the conversions in Table 7.

Table 7 converts each annual local-currency observation using the contemporaneous period-average official exchange rate rather than a current exchange rate: MYR/USD 4.561 in 2023 and 4.576 in 2024, and VND/USD 23,787.3 in 2023 and 24,164.9 in 2024. The convention is the World Bank World Development Indicators series PA.NUS.FCRF, sourced from the IMF International Financial Statistics database (World Bank n.d.). Under that convention, Malaysia rises from about US$252.1 million to US$458.0 million (+US$205.9 million), while Vietnam rises from about US$289.9 million to US$359.5 million (+US$69.6 million). The difference between those changes is about US$136.3 million.

Table 7. Malaysia's 2024 base expansion: descriptive collection benchmarks using historical period-average USD conversion

Note: Table values are reported in U.S. dollars and are rounded from contemporaneous annual period-average conversions; no current or 2026 exchange rate is used. Source-of-record amounts are RM1.15B (2023 SToDS), RM1.62B (2024 SToDS), and RM476M (2024 LVG) for Malaysia, and VND6.896T (2023) and VND8.687T (2024) for Vietnam's foreign-supplier portal.

Series	2023	2024	Change
Malaysia (SToDS + LVG)	$252M	$458M	+$206M
Vietnam (foreign-supplier e-portal)	$290M	$359M	+$70M
Difference between changes	—	—	$136M
Malaysia LVG (USD; RM476M source)	—	$104M	—

The approximately US$104 million directly reported LVG collection (RM476 million as reported) and the approximately US$136 million before-and-after difference are not alternative estimates of one common treatment effect; they are descriptive benchmarks derived from different quantities.

The comparative difference is larger than the converted LVG collection because it also incorporates growth in the pre-existing SToDS stream. Under the same historical-FX convention, SToDS rises from about US$252 million in 2023 to about US$354 million in 2024, while Vietnam's direct-portal series rises from about US$290 million to about US$359 million. Whether the relative acceleration in Malaysia's existing tax stream is attributable to the base expansion or to independent growth cannot be resolved with these data.

On inference. An earlier draft used linearly interpolated quarterly values to report a statistically significant difference-in-differences estimate. Because interpolation mechanically smooths the series and the genuine annual panel contains only four observations, that specification cannot support meaningful inference and no significance claim is made here. Appendix A documents the withdrawn estimate and correction history.

Limitations. Two countries, two years, no pre-trend test. More fundamentally, the two series do not tax the same activity: Malaysia's LVG regime is a sales tax on imported physical goods below a statutory threshold of RM500 (about US$109 at the 2024 period-average USD/MYR rate), while Vietnam's foreign-supplier portal collects VAT and corporate income tax on nonresident digital services. Vietnam's suitability as a comparison therefore rests on continuity of its own portal-based tax mechanism over the period, not on the two tax bases being identical, and the comparison should be read as descriptive rather than identifying the effect. A synthetic control across more ASEAN comparators, a placebo-date test, or a control series drawn from a goods-import regime would each materially strengthen identification in future work.

7. Regional calibration

Table 8 applies the Indonesia-focused base estimate's 5.88% blended retained share to the latest reported digital-economy GMV of Vietnam, the Philippines, Thailand, and Malaysia. These four rows are scenario calibrations rather than independent country-level measurements.

Table 8. ASEAN-5 scenario calibration using the Indonesia-focused base estimate's 5.88% blended retained share

Country	Digital GMV	Platform revenue	Unbooked GTV	Ratio
Indonesia	$71.0B	$4.18B	$66.8B	16.0×
Vietnam	$36.0B	$2.12B	$33.9B	16.0×
Philippines	$31.0B	$1.82B	$29.2B	16.0×
Thailand	$46.0B	$2.70B	$43.3B	16.0×
Malaysia	$31.0B	$1.82B	$29.2B	16.0×
ASEAN-5	$215B	$12.64B	$202B	16.0×

The four non-Indonesian rows apply the same 5.88% retained share. Their identical ratios are arithmetic, not an empirical cross-country regularity.

The base estimate is grounded in platform-level data focused on Indonesia, but the GoTo component is Group-level rather than perfectly country-isolated. Malaysia alone has an independent policy benchmark (Section 6). The regional dollar total should therefore be read as an order-of-magnitude scenario anchored on the Indonesia-focused base rather than as evidence of a common cross-country ratio.

This calibration is not a single-year, single-scope regional estimate, and should not be read as one. The $71.0B base enters Table 8 from the 2023 three-platform construction in Table 3; Grab and Shopee are Indonesia-specific or derived for Indonesia, while GoTo is Group-level. The other four countries enter at their 2024 economy-wide digital GMV, since no comparable three-platform figure exists for those markets.

The most directly supported headline magnitude is the $66.8 billion Indonesia-focused estimate in Table 3; the ASEAN-5 total is an order-of-magnitude scenario built on that base, not an additional independently measured observation.

7.1 Cross-check against published tax collections

To place the calibrated wedge magnitudes in fiscal context, Table 9 compares them with published 2024 digital-tax collections for the three countries with available collection totals, converting local-currency figures at each country's 2024 period-average official exchange rate.

Table 9. Published 2024 digital-tax collections as a share of calibrated unbooked GTV

Country	Unbooked GTV	Tax revenue	Collections / calibrated wedge
Indonesia	$66.8B	$0.749B	1.12%
Malaysia	$29.2B	$0.458B	1.57%
Vietnam	$33.9B	$0.359B	1.06%

These are descriptive ratios across differently defined tax bases; they are not validation tests of the full regional estimate. Source amounts are Rp11.87T for Indonesia, RM2.096B for Malaysia (SToDS + LVG), and VND8.687T for Vietnam's foreign-supplier portal, converted at 2024 period-average official exchange rates.

8. Do investors respond to the size of the Invisible Ledger?

8.1 Design

If changes in the ecosystem ratio contain incremental information about platform scale or monetization, announcement-window returns may respond when the underlying GMV and revenue figures are released. Earnings announcements are the natural test point because the ratio becomes observable when those inputs are disclosed.

The test is a market-adjusted event study (Brown and Warner 1985; MacKinlay 1997). Abnormal return is the platform’s daily return less the matched market-index return — the NASDAQ Composite for Grab (Nasdaq: GRAB), the Jakarta Composite Index (JCI) for GoTo (IDX: GOTO), and the S&P 500 for Sea (NYSE: SE); cumulative abnormal return (CAR) sums this over the trading day before, of, and after the announcement date. The ecosystem ratio is examined both in levels and as quarter-over-quarter (QoQ) percentage change. Because reporting scope differs across platforms, pooled ratio levels are interpreted only descriptively; QoQ change provides the cleaner within-platform comparison.

A quarter is coded as guidance-contaminated when the same earnings release also revises forward guidance or announces another major concurrent event (merger or acquisition, capital raise, restructuring); these quarters are flagged in the underlying panel and reported both included and excluded, since a same-day guidance change confounds the price reaction the event study is trying to isolate.

The small, unbalanced event sample does not support precise factor-model or subgroup inference. The analysis is therefore treated as exploratory evidence on sign and rough magnitude rather than as a decisive pricing test.

8.2 Results

Table 10. Pearson correlations between the ecosystem ratio and 3-day cumulative abnormal returns

Specification	N	Pearson r	p-value
Level, all observations	35	0.099	0.57
Level, excluding guidance-contaminated quarters	23	0.094	0.67

Specification	N	Pearson r	p-value
QoQ change, all observations	32	−0.011	0.95
QoQ change, excluding contaminated	21	−0.085	0.71
Level, post-structural-break period (2024Q1+)	21	0.112	0.63
QoQ change, post-break period	21	0.294	0.20
Level, post-break and uncontaminated	12	−0.013	0.97
QoQ change, post-break and uncontaminated	12	0.309	0.33

No specification in Table 10 reaches conventional statistical significance; the smallest two-sided Pearson p-value is 0.20. The largest coefficient in magnitude is the QoQ-change correlation restricted to the post-break, uncontaminated subsample: $r = 0.309$ ($n = 12$, $p = 0.33$) — nearly identical to the broader post-break QoQ estimate ($r = 0.294$, $n = 21$, $p = 0.20$) rather than collapsing toward zero the way the level specification does under the same filter (0.112 to −0.013). That stability is not evidence of a real relationship, however: a rank-based (Spearman) re-estimation of the same twelve observations gives $\rho = -0.035$ ($p = 0.91$), and dropping the single largest quarter-over-quarter change in the subsample (Sea, 2025Q1, +42.9%) moves the Pearson coefficient to 0.531 ($n = 11$) while the Spearman estimate stays near zero ($\rho = -0.073$). The discrepancy between the Pearson and rank-based estimates, together with the sensitivity to a single observation, indicates that the positive Pearson association is not robust.

Mean 3-day CAR by platform gives a sense of the baseline noise these correlations are being asked to rise above: Grab −0.99% (n=14, sd=8.97%), Sea +0.22% (n=12, sd=17.23%), GoTo +0.38% (n=9, sd=5.18%).
Sea's high announcement-window dispersion illustrates how much idiosyncratic return noise the ratio-based association must overcome in this small sample.

8.3 Reading the null honestly

At n=35, this test cannot distinguish "no relationship" from "not enough data to detect one." Both readings are legitimate, and neither is overstated here. What can be said without qualification: the ecosystem ratio does not show an easily detected announcement-window pricing effect in this sample, across eight different ways of cutting it, none of which is statistically significant at conventional levels.

Given the earlier withdrawal of an over-stated Malaysia specification (Appendix A), the investor result is reported without reinterpretation: the current sample does not support a robust announcement-window pricing claim.

8.4 Two findings that survive the null

The pricing test was not the only thing this panel produced.

Grab's ecosystem ratio has settled into a descriptively stable operating range. It fell from 21.1× in 2022Q1 to roughly 6.3–6.7× by early 2024, and has held inside that narrow band for eight consecutive quarters (2024Q1–2025Q4) since. This pattern is consistent with descriptive stabilization after Grab's early post-listing period, subject to the GMV-definition continuity caveat in Section 8.5. A ratio this stable over two full years is itself informative about how Grab's disclosed-versus-booked gap behaves once transient listing effects wash out, and is worth reporting as a finding in its own right.

GoTo's much higher ratio (mean 38.5×) is a reporting-convention artifact, not evidence of a larger unbooked economy. Section 4.2 of this paper already establishes why: GoTo's headline GTV figure captures Tokopedia's full marketplace transaction value, while only a thin commission is booked as revenue—a different accounting convention from Grab's On-Demand GMV or Sea's Shopee-segment GMV, not evidence of a larger underlying unreported economy. The cross-platform level difference in Table 5 should therefore be read as a reporting-scope difference rather than as a substantive ranking.

8.5 Limitations

Two structural breaks sit inside this panel—GoTo's Tokopedia deconsolidation (2024-02-01) and Grab's 2022–2023 revenue-recognition normalization. Pooling across both regimes, as the "all observations" rows do, is a real compromise; the post-break-only rows are the cleaner test, and none of them is statistically significant either, though the QoQ-change coefficient in that subsample ($r = 0.29$–0.31 depending on the contamination filter) is the largest in the table and is not robust to rank-based re-estimation or to single-observation removal (Section 8.2).

The 3-day window is a first cut. A narrower [0,+1] window was not attempted here and could sharpen the estimate in either direction.

Grab’s GMV terminology changed between 2022–2023 (“Total GMV”) and 2024 onward (“On-Demand GMV”); the two appear to share scope based on one release’s own wording, but this is not independently confirmed across the full sample, so the series may not be perfectly continuous across that boundary.

Coverage remains uneven because an observation is included only when GMV/GTV, revenue, and the announcement date can all be traced to primary public disclosures; quarters lacking one of these inputs are omitted rather than interpolated.

9. Scope, interpretation, and limitations

9.1 GTV is not value-added or tax due

The wedge is gross transaction value not booked as platform revenue. It can include participant receipts, cost of goods, vehicle or labour costs, taxes already paid, refunds, and pass-through amounts. The paper therefore should not convert the full wedge into “untaxed value creation” without participant-level cost and filing data. Estimating value added would require participant-level cost and filing data that are not available in the current disclosures, so this paper does not estimate it.

9.2 What remains outside the estimate

The estimates cover selected transaction platforms and omit many forms of digital labour, creator income, direct social-commerce transactions, and activity on smaller platforms. No separate creator-economy range is added because a source of comparable provenance has not been identified. This omission means the paper is incomplete as a census of platform income, but it also avoids adding a weakly sourced number to a calculation already dependent on calibration.

9.3 Data-provenance risk in the regional total

The four non-Indonesian inputs in Table 8 are 2024 economy-wide digital GMV from Google, Temasek and Bain (2024): $31 billion for Malaysia, $36 billion for Vietnam, $31 billion for the Philippines, and $46 billion for Thailand. These inputs differ in both year and scope from the 2023 three-platform base. The base itself is Indonesia-focused rather than perfectly country-isolated because the GoTo row uses Group-level disclosures. Table 8 therefore remains a calibrated scenario rather than a set of independent country measurements, and the paper's empirical weight rests on the transparent source hierarchy rather than on a claim of uniform geographic scope.

10. Policy implications: reporting first, liability second

The evidence supports a distinction between information collection and tax liability. OECD platform-reporting work and the European Union’s DAC7 regime show that platforms can be required to identify sellers, collect transaction information, and report income-relevant data to tax authorities (OECD 2019, 2020a; European Union 2021). Those reporting duties do not imply that every gross payout should be taxed or withheld at the same rate. They create an auditable record from which thresholds, expenses, residence, and existing obligations can be determined.

For low-income or subsistence participants, an initial zero-liability registration channel could improve administrative visibility without immediately imposing a payment obligation. This is a design hypothesis, not a result established by the current data. Experiments in Sri Lanka and Benin show that formalisation is costly, take-up is limited without substantial incentives, and benefits are heterogeneous (de Mel, McKenzie and Woodruff 2013; Benhassine et al. 2018). Any zero-rated registration proposal should therefore be tested against participation, compliance costs, access to benefits, and the risk of driving activity away from formal platforms.

For participants above a clearly defined threshold, the strongest evidence supports platform-level information reporting. Withholding may be appropriate where the tax base is simple and deductible costs can be handled, but it is not automatically superior in every platform vertical. The platform already computes payouts and maintains identity and transaction records, which lowers reporting cost; whether it should also remit tax depends on local law, participant expenses, and administrative capacity (Kleven, Kreiner and Saez 2016; Slemrod 2019). The policy sequence implied by the paper is therefore: create reliable reporting, distinguish participant types, then evaluate withholding rather than treating it as the default consequence of visibility.

The ratio evidence measures the scale of the administrative problem. It does not determine the correct threshold, rate, deduction rule, or social-protection design. Those parameters require participant-level microdata and policy pilots.

11. Conclusion

This paper estimates a large gap between platform-mediated transaction value and the revenue booked by the platforms that coordinate it. The 2023 three-platform Indonesia-focused estimate is $66.8 billion, while a four-country scenario calibration—applying the base estimate's blended take rate to each country's most recently reported digital-economy size, not a same-year regional measurement—suggests an ASEAN-5 total of approximately $200 billion and an ecosystem ratio of 16.0×. The calculation is most credible as an order-of-magnitude measure of recorded participant-facing transaction value outside routine third-party reporting. It is not a direct measure of profit, unpaid tax, or value-added.

The primary-source GoTo Group time series shows that the share of GTV not booked as net revenue remains between 97.6 percent and 98.9 percent over 2020–2023; the pattern is persistent but not monotonically increasing. Scope-matched historical Grab and Shopee disclosures likewise show large transaction-to-revenue gaps, although levels vary with monetization and accounting scope. Malaysia's policy expansion provides a limited descriptive benchmark—about US$104 million of directly reported 2024 LVG collections (RM476 million as reported) and an approximately US$136 million before-and-after difference against Vietnam under contemporaneous historical FX—but the unmatched tax bases and four-observation panel rule out a causal interpretation. The regional calibration remains a scenario, and the exploratory investor analysis likewise does not show a stable announcement-window pricing relationship in the current sample.

The policy implication is consequently modest but actionable. Platforms already generate the information that modern tax systems normally obtain from fiscal intermediaries. Reporting rules can convert that private record into administrative visibility. Liability, withholding, deductions, and formalisation benefits should then be designed separately rather than inferred from gross transaction value.

Appendix A. Data provenance and robustness notes

This paper has undergone five documented correction rounds. Each round is recorded here so that the body of the paper can be read as an argument rather than as an errata list.

First round: sourcing. An earlier version cited two of three Indonesia platform figures to SEC filings that do not contain those figures at the country level. Grab's Indonesia revenue is $605 million as directly disclosed in its 20-F, not the $1.2 billion previously reported; Sea Limited's 20-F does not disclose an Indonesia-specific figure for Shopee at all, only a combined Southeast Asia total. Both are now labelled by provenance in Table 2.

Second round: ratio definition and inference. The ecosystem ratio was reported throughout as GTV ÷ Revenue while being defined as (GTV - Revenue) ÷ Revenue. The underlying data files compute it correctly under the stated definition; the error was introduced in the paper's tables. Every ratio is accordingly reduced by exactly one: 17.0× becomes 16.0× for both Indonesia and the ASEAN-5. Dollar magnitudes are unaffected, since the wedge itself was always computed as GTV - Revenue. Separately, the Malaysia difference-in-differences estimate was previously derived from an interpolated quarterly panel and reported with a p-value below .001; that claim is withdrawn, for the reasons given in Section 6.

Third round: provenance and figure values. The introduction's opening figure for Southeast Asia's 2024 digital-economy size was reported as $200 billion; the figure in the cited report (Google, Temasek and Bain 2024) is $263 billion, which is also the figure the January 2026 SSRN posting used. The error was introduced during revision and is now reverted. Separately, Figure 2's "original draft" bars previously showed $225 billion and 12.4×, values matching no version of this paper; the SSRN posting reports $192 billion and 12.3×, and the figure now shows those.

Fourth round: country volume verification. Every calibrated country volume in Table 8 has now been checked against the page of the cited source rather than against the paper's internal arithmetic. All four were wrong: Malaysia was recorded as $16.0B against a reported $31B, Thailand as $24.0B against $46B, the Philippines as $38.0B against $31B, and Vietnam as $32.0B against $36B. The Malaysia figure corresponds exactly to the e-commerce sub-total shown on the same page and appears to have been taken from that sub-chart rather than from the overall digital-economy figure. Two of the four errors, Malaysia and the Philippines, were introduced after the January 2026 SSRN posting, which carried the correct values. Correcting all four raises the ASEAN-5 calibrated total from $170 billion to approximately

$200 billion and narrows the then-reported effective-rate range in Table 9 from 1.23-3.34% to 1.11-1.72%. At that stage, the headline base estimate, 16.0× ratio, GoTo time series, and Malaysia–Vietnam comparison were unaffected by the country-volume correction; the fifth-round source-lineage audit below subsequently revisits the GoTo and tax-collection series independently.

Residual rounding. The Indonesia and ASEAN-5 ratios were reported as 15.9× and 16.1× respectively while the four calibrated countries were reported at exactly 16.0×. Since all five rows are generated by one take rate, they cannot differ; the spread arose from rounding revenue to one decimal place. All ratios are calculated from unrounded source values; displayed components may differ slightly because of rounding.

Historical robustness extension. New primary-source files add scope-matched Grab Mobility + Deliveries GMV/revenue for 2019–2021 and Shopee GMV/e-commerce segment revenue for 2017–2021. These observations are reported separately in Table 3B because they are company/segment-level rather than Indonesia-specific and do not change the Table 3 Indonesia estimate.

Fifth round: currency and source-lineage audit. Every annual local-currency flow is now converted using a contemporaneous period-average official exchange rate (World Bank WDI indicator PA.NUS.FCRF, sourced from IMF International Financial Statistics), rather than a current exchange rate. The audit identified two deeper lineage problems. First, the earlier GoTo 2020–2023 table mixed legacy U.S.-dollar figures with Group disclosures; Table 6 is rebuilt from GoTo's primary pro forma/Group GTV and net-revenue figures in rupiah. The corrected series shows a persistent 97.6–98.9% unbooked share rather than a rise from 95.7% to 97.6%. Second, the Malaysia–Vietnam comparison used legacy dollar conversions not tied to one historical FX rule; Table 7 is restated from RM and VND source amounts and now gives an approximately US$136 million descriptive difference, while Malaysia's directly reported RM476 million LVG collection converts to about US$104 million. Table 9 is likewise restated from 2024 local-currency collections. GoTo's 2023 Group GTV and net revenue still convert to approximately US$39.8 billion and US$0.97 billion, so the $66.8 billion headline magnitude and 16.0× ratio are numerically unchanged, but the manuscript now labels the base estimate Indonesia-focused because GoTo's Group disclosure is not a country-level split.

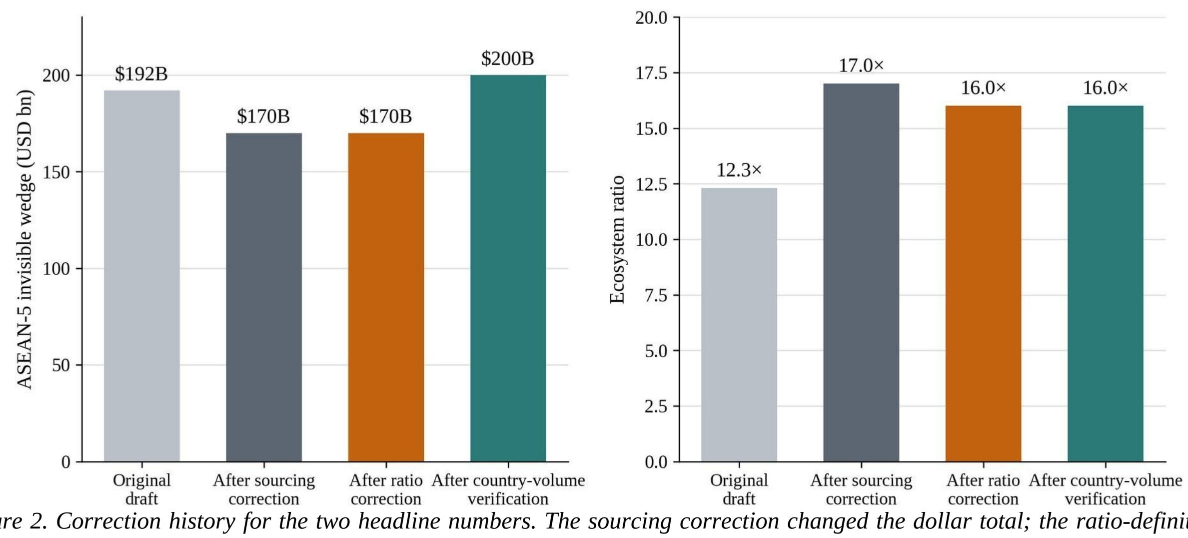

Figure 2. Correction history for the two headline numbers. The sourcing correction changed the dollar total; the ratio-definition correction changed the ratio without changing the wedge; the country-volume verification raised the dollar total again without changing the ratio. The fifth-round currency/provenance audit changes supporting series and scope labels but leaves the $66.8B headline wedge and 16.0× ratio numerically unchanged.

Appendix B. Variable definitions and data sources

Gross transaction value (GTV) / gross merchandise value (GMV). The total value of transactions processed through a platform in a period, regardless of who ultimately receives the proceeds; company filings use both terms depending on platform and segment.

Platform revenue / net revenue. Revenue recognized by the platform on its own income statement from commissions, fees, and directly billed services; it excludes participant-facing pass-through value not booked as platform revenue.

Unbooked GTV. GTV minus platform revenue. This is computed from disclosed or transparently derived inputs and is not separately reported by the platforms.

Ecosystem ratio. Unbooked GTV divided by platform revenue, or (GTV − revenue) / revenue. It measures participant-facing transaction value relative to booked platform revenue and is not a measure of profit, tax liability, or value-added.

Take rate. Platform revenue divided by GTV.

Blended take rate. The combined take rate computed across the three rows in the 2023 Indonesia-focused base estimate: total platform revenue divided by total GTV (5.88%). Because the GoTo input is Group-level, this is not a perfectly country-isolated Indonesia parameter. It is used only for the four-country scenario calibration in Section 7.

Digital GMV. Country-level 2024 digital-economy gross merchandise value reported by Google, Temasek and Bain (2024) for the four non-Indonesian markets in Table 8.

Cumulative abnormal return (CAR). The platform's return less its matched market-index return, cumulated over the event window around an earnings announcement.

Guidance-contaminated event. An earnings announcement that also revises forward guidance or announces a major concurrent corporate event, such as a merger or acquisition, capital raise, or restructuring.

Low Value Goods (LVG). Malaysia's 2024 sales-tax regime for imported low-value goods used in Section 6 as a descriptive policy benchmark.

Sales Tax on Digital Services (SToDS). Malaysia's pre-existing digital-services tax stream, reported separately from LVG collections.

Jakarta Composite Index (JCI). The Indonesian equity-market benchmark used for GoTo in the event-study return matching.

NASDAQ Composite Index. The U.S. equity-market benchmark used for Grab in the event-study return matching.

S&P 500. The U.S. equity-market benchmark used for Sea in the event-study return matching.

Central Index Key (CIK). The unique identifier assigned by the U.S. Securities and Exchange Commission to a filer and used to retrieve Grab and Sea filings from data.sec.gov.

FX conversion convention. Annual monetary flows reported only in local currency are converted to U.S. dollars using the observation year's official exchange rate (local-currency units per US$, period average), World Bank WDI indicator PA.NUS.FCRF, sourced from the IMF International Financial Statistics database (World Bank n.d.). Ratios are calculated in the source currency before conversion whenever possible; legal thresholds retain their original currency with any USD equivalent labelled as a conversion.

Data statement

Primary platform inputs are drawn from company filings and releases. Grab and Sea Limited data are taken from U.S. Securities and Exchange Commission filings on data.sec.gov; the historical robustness files add Grab segment and geographic revenue for 2019–2021 and Shopee GMV and e-commerce segment revenue for 2017–2021. GoTo Table 6 uses the company's FY2022 and FY2023 annual reports: 2020–2021 are pro forma Group figures and 2022–2023 are reported Group figures; these are not treated as country-isolated Indonesia observations. The 2022Q1 observation retained in the separate investor-panel source-lineage file is derived as first-half 2022 minus directly reported 2022Q2 and is not treated as a standalone public-announcement observation. LSEG/Refinitiv pulls are used only as external cross-checks, not as the source of record. Malaysian collection inputs are RM1.15B SToDS (2023), RM1.62B SToDS (2024), and RM476M LVG (2024) from parliamentary reporting. Vietnam's direct foreign-supplier portal inputs are VND6.896T (2023) and VND8.687T (2024) from tax-authority reporting. Indonesia's Table 9 input is the Directorate General of Taxes' Rp11.87T 2024 digital-sector collection total. Annual local-currency amounts are converted with contemporaneous World Bank WDI/IMF period-average official exchange rates. Code and source-lineage files are available from the author upon request.

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